Composite Pattern

Architecture of Computer Games
SE 456

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11 February 2019

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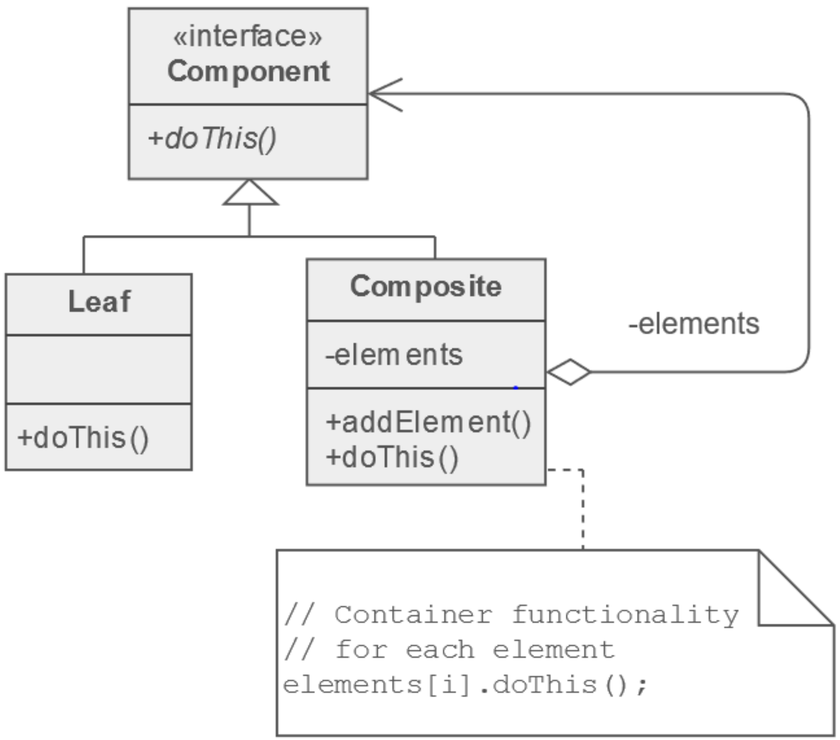
Composite

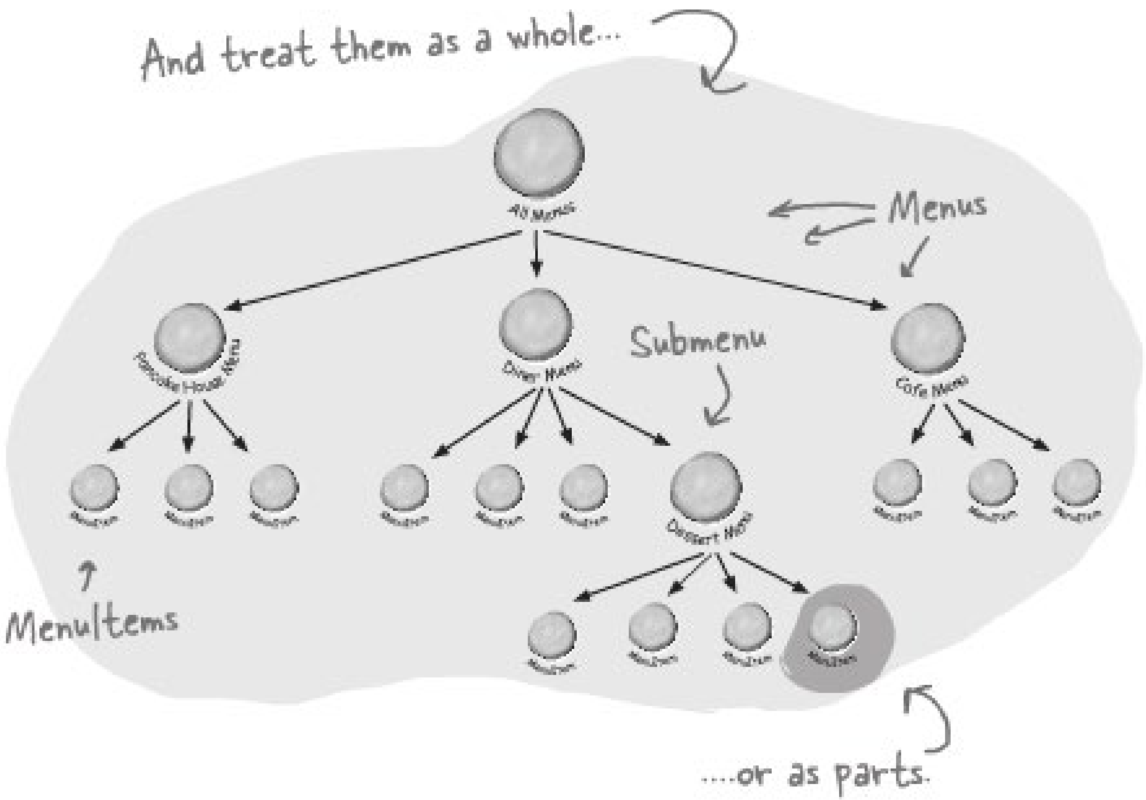
- Intent
  - Compose objects into tree structures to represent whole-part hierarchies.
  - Composite lets clients treat individual objects and compositions of objects uniformly.
- Remarks
  - Recursive composition
  - 1-to-many "has a" up the "is a" hierarchy



Composite

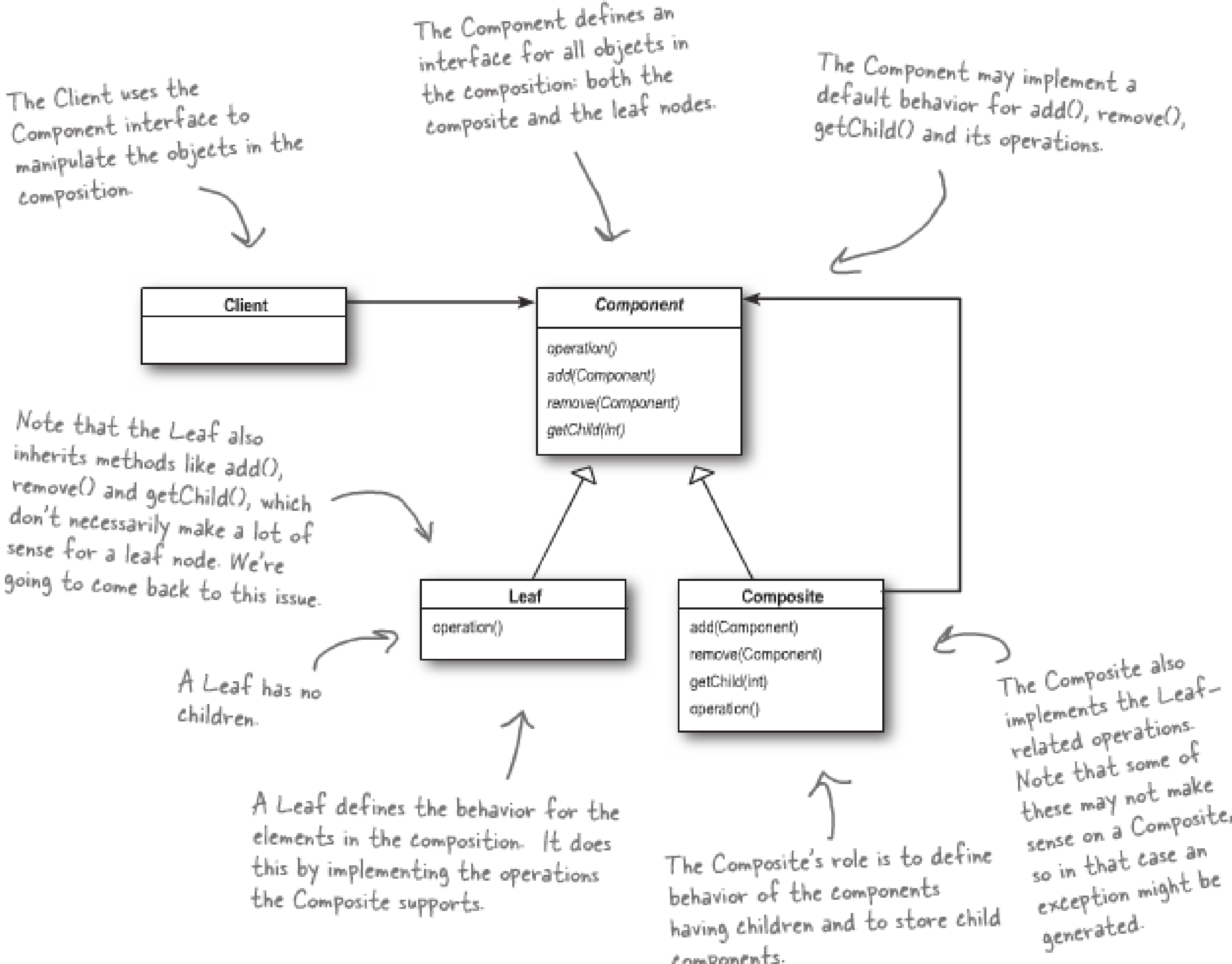
- Composite Pattern allows you to compose objects into tree structures to represent part-whole hierarchies.
- Composite lets clients treat individual objects and compositions of objects uniformly.

Composite



And treat them as a whole...
Menus
Submenu
MenuItems
....or as parts.

Composite

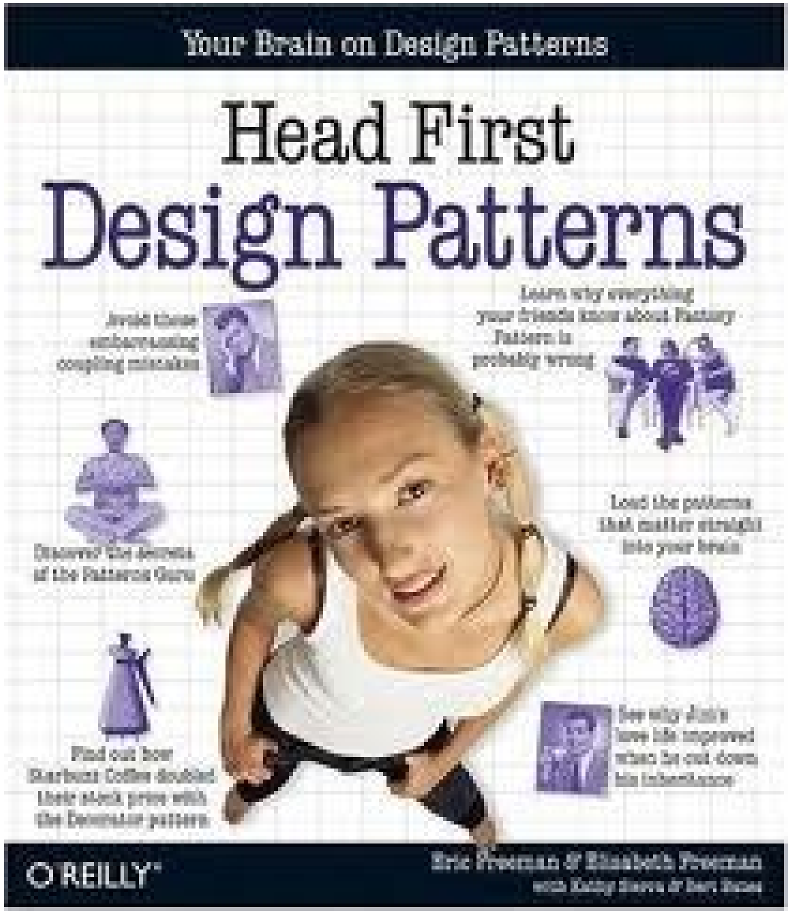

The Client uses the Component interface to manipulate the objects in the composition.
The Component defines an interface for all objects in the composition: both the composite and the leaf nodes.
The Component may implement a default behavior for add(), remove(), getChild() and its operations.
Client
Component
operation()
add(Component)
remove(Component)
getChild(int)
Note that the Leaf also inherits methods like add(), remove() and getChild(), which don't necessarily make a lot of sense for a leaf node. We're going to come back to this issue.
Leaf
operation()
Composite
add(Component)
remove(Component)
getChild(int)
operation()
A Leaf has no children.
A Leaf defines the behavior for the elements in the composition. It does this by implementing the operations the Composite supports.
The Composite's role is to define behavior of the components having children and to store child components.
The Composite also implements the Leaf-related operations. Note that some of these may not make sense on a Composite, so in that case an exception might be generated.

Composite

- Defines class hierarchies consisting of primitive objects and composite objects
  - Primitive objects can be composed into more complex objects
- Can treat primitive and composite objects uniformly
- Adding new components should be easy
  - Client code does not need to be changed
    - Uses only the component interface.

Design Patterns

- Easy tutorial book on design patterns
  - Get your copy today
  - Ecopy at Depaul Library through the safari link



Thank You!

- Questions?

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